APPLIED CHEMISTRY

1. Write the flow sheet diagram for the production of cement.
2. Differentiate between OPC and PPC cement.
3. Write the flow sheet diagram for paper production.

Write monomers and one use of:

- Bakelite
- Nylon-6,6
- polyvinyl chloride(PVC)
- Teflon

Write an example and one use of:

- Analgesic, antipyretic, antibiotics, and antiseptic drugs.
- Herbicide and Pesticide
- Germicide and Insecticide
- A synthetic fertilizer – nitrogen fertilizer/Phosphorus (phosphatic) fertilizer, mixed(NPK) fertilizer.

7. Why is chlorobenzene o/p – directed towards an electrophilic substitution reaction?
8. It is dangerous to boil a sample of ether stored for a long time, give a reason.
9. Ether is stored in a bottle containing iron wire, why?
10. Give a suitable test to distinguish ethanamine from N-methyl methenamine.
11. Write a chemical test to distinguish ethanoic acid(acetic acid) from methanoic acid(formic acid).
12. Why is a chloroacetic acid stronger acid than acetic acid?
13. Why is a formic acid stronger acid than acetic acid?
14. Why an amino group of aniline is protected before nitration? (Aniline can not be nitrated directly, why?)
15. Write the functional isomers of C3H6O with their IUPAC name. Give a chemical test to distinguish them.
16. Write an unsymmetrical ether of C3H8O. How would you prepare this ether using Williamson's synthesis?
17. Write down possible isomeric amines of C3H9N and give their IUPAC names.

What happens when:

1. Sodium benzoate is heated with soda lime.
2. Phenol is heated with zinc dust.
3. Chlorobenzene is treated with chloral.
4. Ethyl alcohol(ethanol) is treated with acetic acid(ethanoic acid)
5. Phenol is treaded with aq. Br2.
6. Aniline is treated with aq. Br2.
7. Phenol is treated with benzene diazonium chloride.
8. Ethoxy ethane is treated with excess HI.
9. Methanal(formaldehyde) is treated with ammonia.
10. Methanal/benzaldehyde is treated with NaOH.
11. Ethanal/propanone is treated with NaOH.
12. Aldehyde/ketone is treated with hydroxyl amine.
13. Ethanol is heated with conc. H2SO4.
14. Acetic acid is treated with P2O5.

5. All reactions of chloroform.

SET 6: Reactions to prepare: (with one use)

1. DDT
2. Picric acid
3. BHC
4. Urotropin
5. Chloropicrin
6. TNT

SET 7 : Convert:

1. 1-bromopropane to 2- bromopropane and vice-versa.
2. 1- propanol to 2- propanol and vice versa.
3. Methenamine to ethanamine and vice versa.
4. Phenol to anisole(methoxy benzene) and vice versa.
5. Ethoxy ethane to methoxy ethane.
6. Phenol/aniline to azo-dye.
7. Ethanal/ ethanol to 3-hydroxy butanal.
8. Ethanol to 2- hydroxy propanoic acid.
9. Propanone(acetone) to 2-hydroxy-2-methyl propanoic acid.
10. Phenol to toluene.
11. Ethanol to propanol/ propanoic acid.
12. Methanamide to ethanamine.

Some Important questions of Organic Chemistry:

1. Why is chloroform stored in a dark bottle containing ethanol?
2. Why does chloroform not give white ppt. with aq. $AgNO_3$?
3. Why is nucleophilic substitution reaction difficult in haloarene?
4. Why is the boiling point of ethanol greater than that of ethoxy ethane?
5. Why is phenol more acidic than aliphatic alcohol?
6. Why does nitrobenzene undergo an electrophilic substitution reaction at the meta position? (Explain why –the NO_2 group is meta-directing towards electrophilic aromatic substitution)

- Identification of 1, 2, and 3 alcohols by Victor Meyer's method.
- Separation of 1, 2, and 3 amines by Hoffmann's method.

SET 3: Name reactions:

1. Aldol / cross aldol condensation reaction
2. Cannizaro's reaction
3. Perkins condensation
4. Wolff kishner reduction
5. Cannizzaro reaction
6. Claisen Condensation reaction.
7. Benzoin condensation
8. Carbylamine reaction
9. Reimer Tiemann reaction
10. Coupling reaction(preparation of azo dyes)
11. Diazotization reaction
12. Friedel- Craft's reaction
13. Hoffmann's Bromamide reaction(De carbonylation reaction)
14. Esterification reaction
15. Carbonylation reaction (Oxo process)
16. Williamson's etherification reaction
17. Rosenmund reduction

SET 4: Test reaction:

1. Iodoform test
2. Silver mirror test(reaction with Tollen's reagent)
3. Nitrous acid test of 10, 20, and 30 amines.
4. 2,4 – DNP test
5. Carbylamine test (test of primary amines)

SET 5: Important reactions:

1. Reduction of nitrobenzene in a different medium.
2. Preparation of alcohols using a Grignard reagent.
3. Oxidation of alcohols.
4. Fermentation

2. Formation of complexes
3. Reason for colored complexes
4. Catalytic property
5. Magnetic property.
6. CFT for the octahedral complex.

SET 2: Extraction of :

1. (Blister) copper from copper pyrites.
2. Steel (Mainly Open Hearth process)
3. Iron from iron pyrites.
4. Zinc from zinc blend (sulphide ore).
5. Mercury from cinnabar (HgS) ore.

SET 3: Chemistry of:

1. Blue vitriol ($CuSO_4.5H_2O$)
2. White vitriol ($ZnSO_4. 7H_2O$)
3. Zinc white (ZnO)
4. Calomel (Hg_2Cl_2)
5. Corrosive sublimate ($HgCl_2$)
6. Rusting of iron
7. BOP

ORGANIC CHEMISTRY

SET 1: preparation of (Reaction only) :

- Chloroform (trichloro methane)
- Nitrobenzene
- Formic acid (methanoic acid)
- Aniline
- Diethyl ether (Ethoxy ethane)

SET 2: Almost one sure :

Unit:5. Electrochemistry

1. What is Electro Chemistry?
2. What are Electrodes?
3. Standard electrode(redox) potential
4. Hydrogen electrodes
5. Calomel Electrodes
6. Electrochemical series
7. Application of Electrochemical series
8. Standard Hydrogen Electrode
9. Cell potential and free energy change
10. What is a cell? Explain its types.
11. Commercial cell

Some Important Questions of Physical Chemistry for Class 12 Chemistry

- Selection of indicators.
- Different concepts of acids and bases and their limitations
- Common ion effect and solubility product principle and its application salt analysis (precipitation reaction).
- Buffer solution.
- Faraday's laws of electrolysis.
- Standard electrodes.
- Hess law.
- Prediction of feasibility/ spontaneity of reactions in terms of free energy change, entropy change, and enthalpy change.
- Differences between order and molecularity of a reaction.
- Collision theory of reaction rate.
- Factors affecting rate of reaction.
- Derivation of integrated rate equation and half-life for a first-order reaction.

INORGANIC CHEMISTRY

SET 1: Characteristics of Transition metals:

1. Variable oxidation state

Explain these terms:

- Rate of reaction
- Rate equation
- Rate constant
- Activation energy
- Effective collision
- Proper orientation

Unit:4. Thermodynamics

1. Define thermodynamics
2. What is the physical concept of entropy?
3. First law of thermodynamics
4. Second law of thermodynamics
5. Third law of thermodynamics
6. State and Explain Hess's law of constant heat summation.
7. Predict the criteria of spontaneity in light of entropy change.

Explain the terms:

- Entropy
- Enthalpy
- Gibb's free energy
- Spontaneous process
- Non-spontaneous process
- Exothermic process
- Endo thermic process
- Isothermal Process
- Adiabatic Process
- Isobaric Process
- Isochoric Process
- Cyclic process

7. Define Lewis Acid and Base with Examples?
8. Write Limitation of Lewis Acid and Lewis Base?
9. Define the Common ion effect?
10. Application of Common ion effect?
11. Solubility product principle and its application.
12. What is the ionic product of water?

Define:

- Acidic salt
- Degree of ionization

Unit:3. Chemical Kinetics

1. Define chemical kinetics.
2. Define the rate of Chemical reaction.
3. Define the half-life of a reaction.
4. Factors affecting the rate of reaction.
5. Write rate law.
6. What is the rate constant? Write its unit?
7. What do you mean by the order of a reaction?
8. Define the first (1st) order reaction with its unit?
9. What is meant by Instantaneous rate of reaction?
10. Differentiate between the order and molecularity of a reaction?
11. How do concentration, temperature, Catalyst, and surface area of reactants affect the rate of reaction?
12. What is meant by the effective collision of reacting species? What are the essential conditions for the effective collision of reacting species?
13. Find the unit of rate constant of zero, first, and second order reaction.
14. Arrhenius equation
15. Catalyst and catalysis
16. Homogenous catalysis
17. Heterogeneous Catalysis
18. Enzyme
19. Explain the role of catalyst in the reaction mechanism

PHYSICAL CHEMISTRY

Unit:1. Volumetric Analysis

1. What are primary and secondary standard solutions?
2. What are the requisites for a substance to be a primary standard?
3. Define molarity and normality. Write their relationship.
4. What is meant by redox titration?
5. How is a suitable indicator selected for a particular titration?

Define the terms:

- Normality
- Molarity
- Normality factor
- Endpoint and Equivalent point
- Titration error
- Semi-normal solution and Decinormal solution
- Indicator
- Alkalimetry
- Acidimetry
- Basicity of acid
- Acidity of base
- Gram equivalent weight

Unit:2. Ionic Equilibrium

1. State Ostwald's dilution law. What is the limitation of this law?
2. Define pH and pOH. Write their relation.
3. What will happen when HCl gas is passed over a saturated solution of NaCl, also Explain the principle involved?
4. Acid and Base
5. Bronsted Acid and Bronsted Base.
6. Conjugate acid and conjugate Base pair of NH3?